

Outline :

1. Review of basic distributions
2. Review of expectation
3. Exercise from textbook.

		Mean	Variance
1. Bernoulli	Ber(p).	p	$p(1-p)$
Binomial	Bin(n, p)	np	$np(1-p)$
Poisson	$\mathcal{P}(\lambda)$	λ	λ
Uniform	$\mathcal{U}([a, b])$.	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential	$\mathcal{E}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$N(\mu, \sigma^2)$	μ	σ^2
Chi-square	χ_p^2	p	
(*) Gamma	Gamma(α, β)		
student-t	t_p		
F	$F_{p,q}$		

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(0, 1)$$

$$Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} N(0, 1)$$

$$X^2 = \sum_{i=1}^n X_i^2 \sim \chi_n^2$$

$$T = \frac{Y_1}{\sqrt{X_n^2}} \sim t_n.$$

$$Y^2 = \sum_{i=1}^m Y_i^2 \sim \chi_m^2$$

$$F = \frac{X^2}{Y^2} \sim F_{n,m}$$

2. Expectation of RV.

Assume X is continuous with pdf $f(x)$.

$$\text{Def: } \mathbb{E}X = \int_{\mathbb{R}} x f(x) dx, \quad \text{Var } X = \mathbb{E}(X - \mathbb{E}X)^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$

$$\text{Prop: } ① \mathbb{E}(X+Y) = \mathbb{E}X + \mathbb{E}Y$$

$$② \mathbb{E}(aX) = a\mathbb{E}X, \quad \forall a \in \mathbb{R}.$$

$$③ \text{ If } X \perp Y, \quad \text{Var}(X+Y) = \text{Var } X + \text{Var } Y.$$

$$④ \text{Var}(aX) = a^2 \text{Var } X$$

✧ e.g. $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2), \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$

$$\text{then } \mathbb{E}\bar{X} = \mathbb{E} \frac{X_1 + \dots + X_n}{n} = \mu.$$

$$\text{Var } \bar{X} = \text{Var} \left(\frac{X_1 + \dots + X_n}{n} \right) = \frac{\sigma^2}{n}.$$



A little bit more of Normal. Let $X \sim N(\mu, \sigma^2)$.

$$\text{Then } \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Besides. Coro: $X \sim N(\mu_1, \sigma_1^2), \quad Y \sim N(\mu_2, \sigma_2^2), \quad X \perp Y,$

$$\text{then } X+Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

3. 7.15 Suppose that $X_1, X_2, \dots, X_m$ and $Y_1, Y_2, \dots, Y_n$ are independent random samples, with the variables X_i normally distributed with mean μ_1 and variance σ_1^2 and the variables Y_i normally distributed with mean μ_2 and variance σ_2^2 . The difference between the sample means, $\bar{X} - \bar{Y}$, is then a linear combination of $m + n$ normally distributed random variables and, by Theorem 6.3, is itself normally distributed.

- a Find $E(\bar{X} - \bar{Y})$.
- b Find $V(\bar{X} - \bar{Y})$.
- c Suppose that $\sigma_1^2 = 2$, $\sigma_2^2 = 2.5$, and $m = n$. Find the sample sizes so that $(\bar{X} - \bar{Y})$ will be within 1 unit of $(\mu_1 - \mu_2)$ with probability .95.

$$\bar{X} \sim N(\mu_1, \frac{\sigma_1^2}{m}) \quad \bar{Y} \sim N(\mu_2, \frac{\sigma_2^2}{n})$$

$$\text{So } \bar{X} - \bar{Y} \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}) \quad (-\bar{Y} \sim N(-\mu_2, \frac{\sigma_2^2}{n}))$$

$$P(|(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)| < 1) = 0.95.$$

$$\Leftrightarrow P\left(\frac{|(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)|}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} < \frac{1}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}}\right) = 0.95$$

$$Z \stackrel{\text{def}}{=} \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} \sim N(0, 1)$$

$$\therefore \frac{1}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}} = Z_{0.025} = 1.96$$